

10. (g) Implementation Plan: B.Tech. in Biotechnology with Specialization in Regenerative Medicine

Semester – I						
Course Code	Course Title	Hours / Week				
		L	T	P	C	
21LEH101T	Communicative English	2	1	0	3	
21MAB101T	Calculus and Linear Algebra	3	1	0	4	
21PYB101J	Physics: Electromagnetic Theory, Quantum Mechanics, Waves and Optics	3	1	2	5	
21MES102L	Engineering Graphics and Design	0	0	4	2	
21EES101T	Electrical and Electronics Engineering	3	1	0	4	
21CYM101T	Environmental Science*	1	0	0	0	
21PDM101L	Professional Skills and Practices	0	0	2	0	
21LEM101T	Constitution of India	1	0	0	0	
Total Credits					18	

Semester – II						
Course Code	Course Title	Hours / Week				
		L	T	P	C	
21LEH102T	Chinese	2	1	0	3	
21LEH103T	French					
21LEH104T	German					
21LEH105T	Japanese					
21LEH106T	Korean					
21LEH107T	Spanish					
21LEH108T	Russian					
21GNH101J	Philosophy of Engineering	1	0	2	2	
21MAB102T	Advanced Calculus and Complex Analysis	3	1	0	4	
21CYB101J	Chemistry	3	1	2	5	
21BTB105T	Cell Biology	2	0	0	2	
21CSS101J	Programming for Problem Solving	3	0	2	4	
21BTC101T	Biochemistry	3	0	0	3	
21MES101L	Basic Civil and Mechanical Workshop	0	0	4	2	
21PDM102L	General Aptitude*	0	0	2	0	
21GNM101L	Physical and Mental Health using Yoga	0	0	2	0	
21GNM102L	National Service Scheme					
21GNM103L	National Cadet Corps					
21GNM104L	National Sports Organization					
Total Credits					25	

Semester – III						
Course Code	Course Title	Hours / Week				
		L	T	P	C	
21CHS251T	Basic Chemical Engineering	3	0	0	3	
21PDH209T	Social Engineering	2	0	0	2	
21BTC201L	Biochemistry Laboratory	0	0	4	2	
21BTC202T	Microbiology	3	0	0	3	
21BTC203L	Cell and Microbiology Laboratory	0	0	4	2	
21BTC204T	Bioprocess Principles	3	0	0	3	
21BTC205L	Bioprocess Principles Laboratory	0	0	4	2	
21BTC206T	Genetics and Cytogenetics	3	0	0	3	
21LEM201T	Professional Ethics	1	0	0	0	
21PDM201L	Verbal Reasoning	0	0	2	0	
21LEM202T	Universal Human Values-II: Understanding Harmony and Ethical Human Conduct	2	1	0	3	
Total Credits					23	

Semester – IV						
Course Code	Course Title	Hours / Week				
		L	T	P	C	
21CHS252J	Chemical Engineering Principles	3	0	2	4	
21CSC206T	Artificial Intelligence	3	0	0	3	
21BTC207T	Molecular Biology	3	0	0	3	
21BTC208L	Molecular Biology Laboratory	0	0	4	2	
21BTC209T	Bioprocess Engineering	3	0	0	3	
21BTC210L	Bioprocess Engineering Laboratory	0	0	4	2	
E	Professional Elective – I	3	0	0	3	
21DCS201P	Design Thinking and Methodology	1	2	0	3	
21PDM202L	Critical and Creative Thinking Skills	0	0	2	0	
Total Credits					23	

Semester – V						
Course Code	Course Title	Hours / Week				
		L	T	P	C	
21MAB303T	Biostatistics for Biotechnologists	3	1	0	4	
21BTC301J	Gene Manipulation and Genomics	3	0	2	4	
21BTC302J	Immunology	3	0	2	4	
21BTC303T	Protein Engineering	3	0	0	3	
E	Professional Elective – II	3	0	0	3	
O	Open Elective – I	3	0	0	3	
21GNP301L	Community Connect	0	0	2	1	
21PDM301L	Analytical and Logical Thinking Skills	0	0	2	0	
21LEM301T	Indian Art Form	1	0	0	0	
Total Credits					22	

Semester – VI						
Course Code	Course Title	Hours / Week				
		L	T	P	C	
21CSS303T	Data Science	1	1	0	2	
21BTC304T	Animal Biotechnology	3	0	0	3	
21BTC305L	Animal Biotechnology Laboratory	0	0	4	2	
21BTC306T	Plant Biotechnology	3	0	0	3	
E	Professional Elective – III	3	0	0	3	
O	Open Elective – II	3	0	0	3	
21BTP302L	Project	0	0	6	3	
21BTP303T	MOOC	3	0	0	3	
21LEM302T	Indian Traditional Knowledge	1	0	0	0	
21PDM302L	Employability Skills and Practices	0	0	2	0	
Total Credits					19	

Semester – VII						
Course Code	Course Title	Hours / Week				
		L	T	P	C	
21BTC401L	Plant Biotechnology Laboratory	0	0	4	2	
21BTC402J	Bio Separation Technology	3	0	2	4	
E	Professional Elective – IV	3	0	0	3	
E	Professional Elective – V	3	0	0	3	
O	Open Elective – III	3	0	0	3	
21GNH401T	Behavioral Psychology	2	0	1	3	
21BTM191T	Bioethics and IPR	1	0	0	0	
Total Credits					18	

Semester – VIII						
Course Code	Course Title	Hours / Week				
		L	T	P	C	
21BTP401L	Major Project	0	0	30	15	
21BTP402L	Major Project	0	0	20	10	
21BTP403L	Internship#	0	0	10	5	
Total Credits					15	

#Students have to register either 21BTP401L or 21BTP402L and 21BTP403L both in eighth semester